

PROJECT SUTRA — PUNCHY GRAND FINALS SCRIPT

FAST-PACED, WORD-FOR-WORD DEFENSE & DISASTER ROBOTICS PITCH (SLIDES 1 TO 3)

PACE: CRISP & ENERGETIC
TOTAL TIME: ~2 MIN 30 SEC

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:35 (35s)

HOOK: "When a disaster strikes, the first 72 hours decide who lives. But the moment drones enter thick forests or mountain ravines, they go blind."

"Respected jury, we are **Team OFFGRID**, and we built **Project SUTRA** — an autonomous drone swarm that finds survivors in GPS-denied, zero-network disaster zones."

"Instead of relying on single manual drones, SUTRA deploys an intelligent swarm backed by 3 breakthrough pillars:"

50Hz GNC **Decentralized Flight:** Drones coordinate locally without towers or cloud servers.

-5dB JSCC **Neural Video:** Feeds stay connected through mountain ridges without blacking out.

3.59cm GEO **3D Terrain AI:** Pinpoints survivors on steep slopes with centimeter accuracy.

[ACTION]: Point to the 3 telemetry boxes and introduce lead roles (Nikhil: GNC & Comms, Vedanth: Vision, Siva: GCS).

SLIDE 02 The Problem: 4 Critical Failure Voids

00:35 - 01:25 (50s)

HOOK: "Why does current search and rescue fail? Because existing drones break down across 4 fatal engineering voids:"

SUBSYSTEM A **Canopy GPS Drift:** Dense trees block satellite signals. Drones drift and crash into trees. (70% loss in Wayanad).

SUBSYSTEM B **Mountain RF Blackout:** Ridges cut radio line-of-sight. Standard video drops off a digital cliff below 5dB SNR.

SUBSYSTEM C **Flat-Earth AI Error:** 2D drone cameras assume the ground is flat. On steep slopes, coordinates miss by 15 to 30 meters.

SUBSYSTEM D **Manual Pilot Bottleneck:** Needs 1 pilot per drone and 15 to 25 crew members, costing ₹12.5 Lakhs per mission.

[ACTION]: Glance across the 4 problem cards on screen (Wayanad crash, Chamoli cut, Sikkim slope drift, crew bottleneck).

 **HOOK:** "SUTRA solves each void with a hardened, defensible technical moat:"

MOAT A SUTRA-FSD & 3D ORCA: Onboard 50Hz obstacle avoidance. Autonomous flight with zero towers.

MOAT B Deep JSCC Neural Video: Analog neural transmission. Clear video down to -5dB SNR with zero digital cliff.

MOAT C 3D DEM AI Geolocation: Edge YOLOv8 + 3D terrain raycasting on Jetson Orin. **3.59cm accuracy** in <15ms.

MOAT D Pegasus 3D GCS: Offline 3D tactical twin. **1 to 2 operators fly the whole swarm** for just **₹95,000 / sortie**.

"The result? SUTRA scans a square mile in **10 to 18 minutes** — 3 to 4 times faster than traditional teams, saving lives when every second counts."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (3-4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000